

Question

Create a main shell loop, display a prompt, read user input, handle exit conditions, capture keyboard input and manage an input buffer.

```
#include <stdio.h>
#include <string.h>

#define BUFFER_SIZE 1024

int main()
{
    char input[BUFFER_SIZE];

    while (1)
    {
        // Display shell prompt
        printf("mysh> ");
        fflush(stdout);

        // Read keyboard input
        if (fgets(input, BUFFER_SIZE, stdin) == NULL)
        {
            printf("\nExiting shell...\n");
            break;
        }

        // Remove newline character
        input[strcspn(input, "\n")] = '\0';

        // Check for exit condition
        if (strcmp(input, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }

        // Handle empty input
        if (strlen(input) == 0)
        {
            continue;
        }
    }
}
```

Output:

```
mysh> hello
You entered: hello
mysh> ls
You entered: ls
mysh> exit
Exiting shell...
rikki@BOOK-GJU4MP9458:~$ |
```